

Smart Scale Systems

Agency and Chatbot Knowledge Guide

A detailed reference for the agency profile, services, team knowledge, chatbot behavior, and the exact process to keep ScaleBot updated whenever service, team, document, or website content changes.

Prepared for	Smart Scale Systems
Purpose	Agency knowledge base, chatbot maintenance, SEO/content consistency
Primary chatbot	ScaleBot - floating website assistant
Key files	src/scripts/robot-mascot.js, api/chat.js, server.js, src/pages/*.html

AI Model Training

AI Automation

Computer Vision

NLP + LLMs

Data Annotation

Training Data

Custom AI

Executive Summary

Smart Scale Systems is positioned as an AI-first services agency that helps businesses scale with intelligent systems. The website presents the agency as a partner for AI model training, automation, computer vision, NLP, LLM solutions, data annotation, AI training data, and custom AI delivery.

ScaleBot should work as a lightweight sales and support assistant. It should answer questions about the agency, guide visitors toward Services or Contact, and stay aligned with the latest website content. Because the chatbot uses a system prompt inside backend files, every business change should be reflected in both website content and chatbot knowledge.

Core Positioning

- AI services agency established in 2021.
- End-to-end AI lifecycle partner: data creation, annotation, training, evaluation, automation, deployment support, and operational AI systems.
- Built for businesses that need practical AI systems, not only prototypes.
- Primary conversion path: explore services, contact the agency, and start a project with a clear action plan.

Agency Service Portfolio

The service portfolio should be treated as the main source of truth for both SEO and chatbot answers. ScaleBot should be able to briefly explain each service, identify related use cases, and recommend contacting the agency when a visitor describes a real project.

Service	What ScaleBot should know
AI Model Training	Custom model training, dataset preparation, fine-tuning, evaluation, optimization, and deployment support.
AI Automation	AI agents, CRM automation, workflow automation, lead generation systems, and process automation.
Computer Vision	Object detection, image classification, segmentation, OCR, video analysis, and object tracking.
NLP	Text classification, sentiment analysis, NER, intent detection, chatbot training data, search relevance, and language understanding.
LLM Solutions	Prompt engineering, LLM fine-tuning, RLHF, response evaluation, assistants, knowledge systems, and custom LLM pipelines.
Data Annotation	Image, video, text, and audio annotation with bounding boxes, polygons, masks, OCR annotation, and QA review.
AI Training Data	Dataset creation, data collection, synthetic data, curation, and benchmark datasets for ML, CV, NLP, LLMs, and automation.
Custom AI Service	Flexible AI builds designed around a client specific workflow, system, business requirement, or technical environment.

Industries and Delivery Strengths

The website positions Smart Scale Systems as useful across multiple industries. Chatbot answers should avoid unsupported regulated claims, but it can confidently mention the following industries and examples already represented on the site.

Industry	Relevant examples
Healthcare and Medical	Medical imaging AI, clinical NLP, diagnostic tools.
Finance and Fintech	Risk models, fraud detection, document automation.
Retail and E-commerce	Visual search, recommendation systems, demand forecasting.
Automotive and Mobility	Autonomous driving datasets, ADAS systems, sensor fusion.
Manufacturing	Defect detection, predictive maintenance, quality inspection.
Tech and SaaS	AI feature development, NLP integrations, LLM products.
EdTech	Intelligent tutoring, content classification, assessment AI.
Security and Defense	Surveillance systems, threat detection, video analytics.

Trust Signals

- Homepage and contact page currently present 120+ projects delivered.
- Client retention is presented as 98%.
- Average rating is presented as 4.9.
- Contact response expectation is within one business day.

Team Knowledge

ScaleBot should answer team questions using a concise role-based summary. It should not invent beyond what appears on the website.

Name	Role	Chatbot-safe summary
Muhammad Shahryar Lodhi	Founder	Leads marketing, outreach, client acquisition, brand strategy, lead generation, and client relationships.
Muhammad Hashir Lodhi	AI/ML Expert	Leads technical AI/ML direction, AI automation, computer vision, deep learning, and scalable AI systems.
Muhammad Nouman Qadeer	AI Engineer	Builds and supports AI solutions, model development, data processing, ML workflows, and AI integrations.

Name	Role	Chatbot-safe summary
Muhammad Mudassir	AI Data Annotator & Labeling Expert	Handles data annotation, labeling, dataset preparation, bounding boxes, segmentation, and QA.

Current Chatbot Architecture

ScaleBot has two main parts: a browser widget and a backend AI endpoint. The browser widget controls the floating robot UI, message display, keyboard behavior, request state, and API calls. The backend endpoint holds the agency knowledge prompt and sends the conversation to Groq.

Layer	File	Responsibility
Frontend widget	src/scripts/robot-mascot.js	Injects floating robot, chat window, styles, message handling, accessibility controls, and POST requests to /api/chat.
Production API	api/chat.js	Vercel-style serverless chat endpoint. Contains system prompt, validates messages/history, and calls Groq.
Local Express API	server.js	Local /chat and /api/chat routes for Express hosting and development. Should stay aligned with api/chat.js.
Content pages	src/pages/*.html	Actual public website content that users and search engines see. Chatbot knowledge should mirror these pages.

Important Behavior Rules for ScaleBot

- Answer in a friendly, concise tone, usually one to three sentences.
- Use light robot personality occasionally, but do not overdo it.
- When a visitor asks for pricing, timeline, or a custom project plan, direct to the relevant page.
- When a visitor asks what the agency does, summarize the service portfolio.
- Never expose credentials, environment variables, API keys, internal debug info.

How to Update the Chatbot When Website Information Changes

Use this workflow whenever a service, team member, contact detail, page layout, or positioning line changes. The goal is to keep the website, SEO metadata

Step	Action
1. Update the public page first	Edit the relevant page under src/pages/. For example, service changes usually start in src/pages/services.html and the matching service detail page.
2. Update SEO metadata	If the changed content affects search intent, update title, meta description, OG tags, canonical URL, sitemap.xml, and route-aware SEO in src/main.jsx.
3. Update chatbot knowledge	Edit the SYSTEM_PROMPT in api/chat.js and server.js. Add the new service, team member, statistic, response rule, or contact detail in both files.
4. Keep answers short and factual	Write chatbot knowledge as crisp bullets. Avoid long marketing paragraphs because overloaded prompts usually produce weaker answers.
5. Test locally	Run npm.cmd run build, then test direct chatbot questions about services, team, contact, pricing/timeline, and the changed item.
6. Deploy and retest	After deployment, ask the live chatbot the same questions and confirm it gives the updated answer.

Recommended Long-Term Chatbot Content System

Right now the chatbot knowledge is duplicated in `api/chat.js` and `server.js`. That works, but it creates maintenance risk because a future update can be applied to one file and missed in the other. The cleaner long-term setup is to move agency facts into one shared knowledge file and make both endpoints read from it.

Recommended source-of-truth file: `content/chatbot-knowledge.md`

Suggested sections for that file

- Agency overview: establishment year, positioning, core promise, target clients.
- Services: one concise entry per service with use cases and links.
- Team: approved public bios only.
- Industries: industry list and allowed examples.
- Contact rules: email, response time, when to direct users to Contact.
- Do-not-say rules: no fake pricing, no unsupported claims, no private credentials, no invented case studies.
- Recent changes: short changelog so future maintainers know what was updated.

Prompt Update Template

Change type	What to add to chatbot knowledge
New service	Service name, one-line description, main use cases, who should ask for it, service page URL.
Changed service	Replace old wording everywhere. Confirm no old service name remains in prompt or page metadata.
New team member	Name, role, public bio, approved skills. Do not add private details.
Changed contact detail	Update contact page, footer, chatbot prompt, JSON-LD, and email links.
New statistic	Add only if it is accurate and approved. Update homepage/contact trust sections and chatbot prompt together.

QA Questions After Every Chatbot Update

- What services does Smart Scale Systems provide?
- Do you offer AI automation?
- Can you help with computer vision or OCR?
- Who is on your team?
- How can I start a project?
- How fast will you respond?
- Do you provide pricing?
- What changed in the new service, team member, or page?

Expected QA outcome: ScaleBot should answer with the latest approved facts, avoid unsupported or serious business inquiries to the Contact page.

Website and SEO Alignment Notes

Search performance and chatbot accuracy depend on consistent wording. If the agency wants to rank for AI Agency in Pakistan, AI services in Pakistan, or specific service queries, the page copy, metadata, sitemap, and chatbot responses should reinforce the same terms naturally.

- Keep one clear H1 per page and align it with search intent.
- Use service-specific H2 sections and plain language descriptions.
- Keep sitemap.xml updated when adding new live routes.
- Use robots.txt to allow crawlers and reference the sitemap.
- Avoid outdated service names in the chatbot after page updates.
- Make sure visual changes do not reduce mobile readability, contrast, or keyboard access.

Final Maintenance Checklist

- Website page updated.
 - Navigation/footer links checked.
 - SEO title and meta description updated if search intent changed.
 - Open Graph tags updated if social preview changed.
 - sitemap.xml updated if a route changed.
 - api/chat.js SYSTEM_PROMPT updated.
 - server.js SYSTEM_PROMPT updated.
 - npm.cmd run build passes.
 - Chatbot asked at least five QA questions.
 - Live deployment retested after publishing.
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